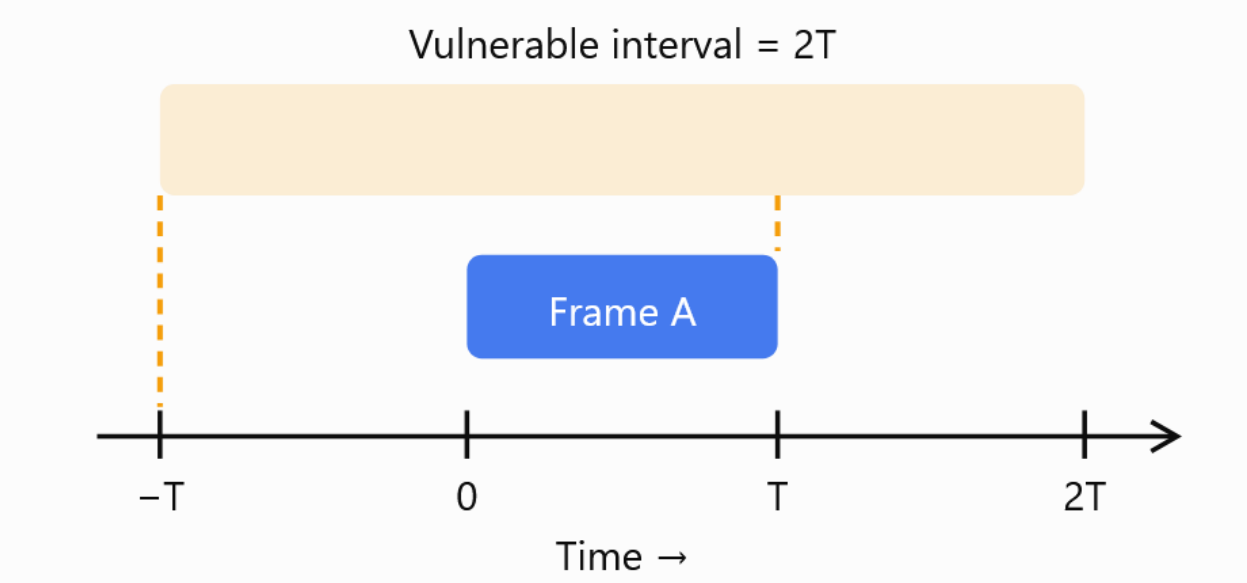
PURE ALOHA THROUGHPUT PROOF

Pure ALOHA is a random-access protocol used in the MAC (Medium Access Control) sublayer. A station can transmit a frame at any time without checking whether the channel is busy.

Symbol	Definition
T	Time required to transmit one frame
λ	Average number of transmission attempts per second
$G = \lambda T$	Average number of transmission attempts per frame time
P_0	Probability of no interfering transmission
S	Normalized successful throughput

Frame transmission attempts follow a Poisson process, frames have equal duration T, and overlapping frames collide.



Another frame starting during the interval from −T to +T can overlap with frame A. Therefore, the vulnerable time is:

$$T_v = 2T$$

For a Poisson process, the probability of k attempts in an interval of length t is:

$$P(N = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

Zero interfering transmission in the in the vulnerable interval. Setting $k = 0$ and $t = 2T$, yield:

$$P_0 = \frac{(\lambda 2T)^0 e^{\lambda 2T}}{0!} \rightarrow P_0 = e^{-\lambda 2T} \rightarrow \textit{we know } G = \lambda T \textit{ then } P_{Success} = e^{-2G}$$

The normalized throughput is the average number of successfully transmitted frames per frame time, therefore

$$S = G \times P_{Success}$$

Substituting the probability of success:

$$S = G \times e^{-2G} \rightarrow S = Ge^{-2G}$$

Then we have $G_{Opt} = \frac{1}{2}$ and $S_{max} = 0.18394 \approx 18.4\ \%$

